

Unit – 3

Sensor, Actuators & Microcontroller

Topic -1- Sensor

A sensor is an electrical device or device that converts a physical quantity (temperature, pressure, humidity, etc.) of the environment into another physical quantity that can be easily understood by humans or machines. In other words: It is an electrical device that detects physical quantities of the environment such as temperature, location, light, pressure, humidity. Etc. and converts it into another electrical signal that is visible on a scale or display by a machine.

Example- Light sensor, pressure sensors, Temperature sensor and many other.

Classification of sensors

Classification Based on Power Supply

1.Passive Sensors:

- **No External Power Source:** These sensors operate without an external power supply.
- **Examples:** Passive infrared (PIR) motion sensors, passive RFID tags.
- **Advantages:** Low power consumption, simplicity.

2. Active Sensors:

- **Require External Power Source:** These sensors need an external power supply to operate.
- **Examples:** Active RFID tags, most electronic sensors.
- **Advantages:** Greater functionality, continuous operation.

3. Self-Powered Sensors:

- **Energy Harvesting:** Generate power from the surrounding environment.
- **Examples:** Solar-powered sensors, piezoelectric sensors.
- **Advantages:** Energy-efficient, sustainable in certain environments.

Classification based on Output

1. Analog sensor –

- Provide a continuous output proportional to the quantity being measured
- **Examples** Analog temperature sensors and Analog pressure sensors.

2. Digital Sensors

- Output discrete, digital signals, often in the form of binary code.
- **Examples** - Digital temperature sensors and digital accelerometers.

Some Most Common Sensors Used in IoT

1. Proximity Sensor:

IoT proximity sensors provide non-contact detection of objects that are enclosed proximity to the sensor. They usually do this by emitting electromagnetic fields or beams of radiation such as infrared. In other words, A device that detects the presence or absence of a nearby object, or properties of that object, and converts it into signal which can be easily read by user or a simple electronic instrument without getting in contact with them.

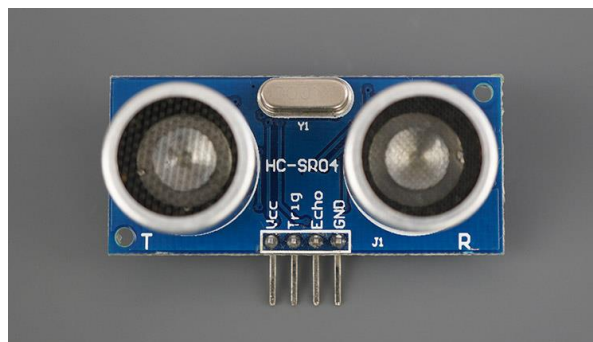
Proximity sensors may also be found in parking lots of malls, stadiums, or airports to indicate available parking spots.



2. Ultrasonic Sensor (HC-SR04):

An ultrasonic sensor is an electronic device that measures the **distance** to an object using ultrasonic sound waves.

It works by emitting an ultrasonic pulse (usually at 40 kHz), then measuring the time it takes for the echo to return after bouncing off a nearby surface. Using this time and the speed of sound, the sensor calculates the distance.



3. LDR (Light Dependent Resistor) :

An **LDR sensor** (Light Dependent Resistor), also known as a **photoresistor**, is a **passive electronic component** that changes its **resistance based on the intensity of light** falling on it. It's one of the most common and simplest light sensors used in electronics and IoT applications.



In **bright light**, the LDR's **resistance decreases** and in **darkness or low light**, the LDR's **resistance increases** .

4. Air Quality Sensor (MQ135) :

The MQ-135 sensor is an air quality sensor used to detect a wide range of gases and pollutants in the environment. It's popular in IoT, smart home, and environmental monitoring projects because it can measure harmful gases commonly found in polluted air. It contains a **chemical-sensitive material** (tin dioxide, SnO_2) that changes **resistance** based on gas concentration.

It detects –

- Carbon dioxide (CO_2)
- Ammonia (NH_3)
- Nitrogen oxides (NO_x)
- Benzene
- Smoke
- Alcohol
- Other volatile organic compounds (VOCs)



According To Our Syllabus We use MQ135 as **Smoke sensor**.

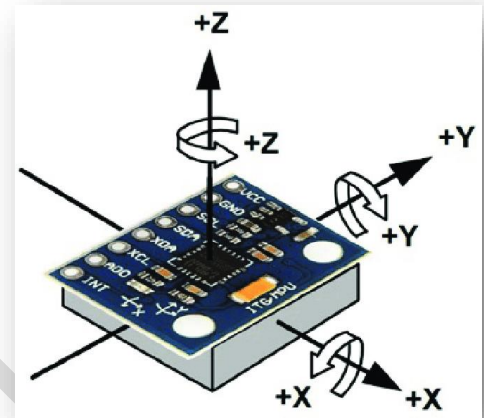
5. Tilt Sensor :

Tilt Sensors are also known as inclinometers. They are a type of position sensor used to measure angles or the slope of an object. It detects the angle of the object from Horizontal or Vertical. This sensor is very much seen in today's smart phones.



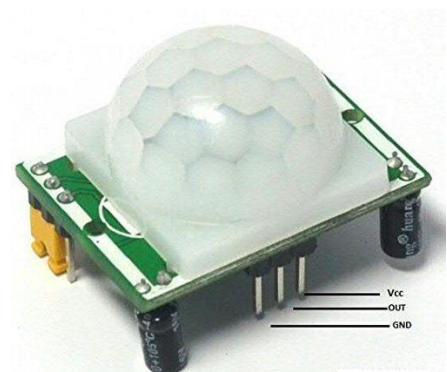
6. Gyroscope Sensor :

A sensor or device which is used to measure **the Angular rate or Angular velocity** is known as Gyro sensors, Gyroscopic sensors ("gyro sensors") are one type of motion sensing device. They are capable of sensing movements that are difficult for humans to detect, such as rotation and changes in orientation in 3 –axis .The most important application is monitoring the orientation of an object. Their main applications are in car navigation systems, game controllers, cellular & camera devices, consumer electronics, robotics control drone & RC control helicopter or UAV control, vehicle control and many more..



7. PIR Motion Detector Sensor:

A passive infrared sensor is an electronic sensor that measures infrared light radiating from objects in its field of view PIR sensors detect motion by sensing infrared radiation emitted by objects with heat, such as humans and animals. These sensors have two slots made of a material sensitive to infrared. When a warm object moves across the sensor's field of view, it causes a change in the infrared levels detected by the two slots. This



difference in infrared radiation is interpreted as motion, triggering the sensor.

8. Touch Sensor:

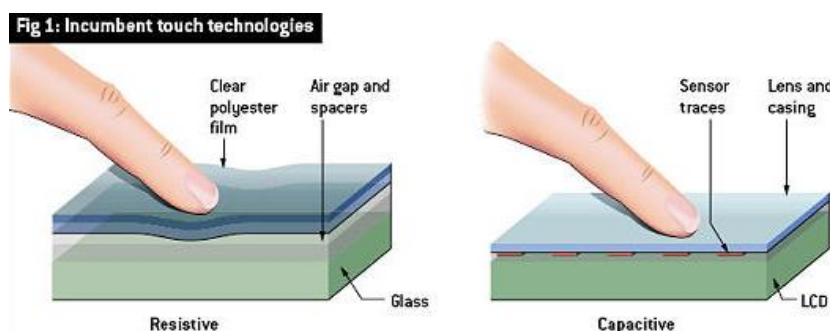
Touch sensor has been common nowadays for displays and lot projects of applications as well. A touch sensor is an electronic sensor used in detecting and recording physical touch. Also known as tactile sensors, it's a small, simple, low-cost sensor. Touch sensor are commonly two types –

1.) Capacitive touch sensor-

Commonly associated with MicroChip AT-42QT1010, a Capacitive touch sensor measures touch based on electrical disturbance from a change in capacitance. It consists of an electrode film on top of the glass panel that's conductively coated with a printed circuit pattern around the outer viewing area.

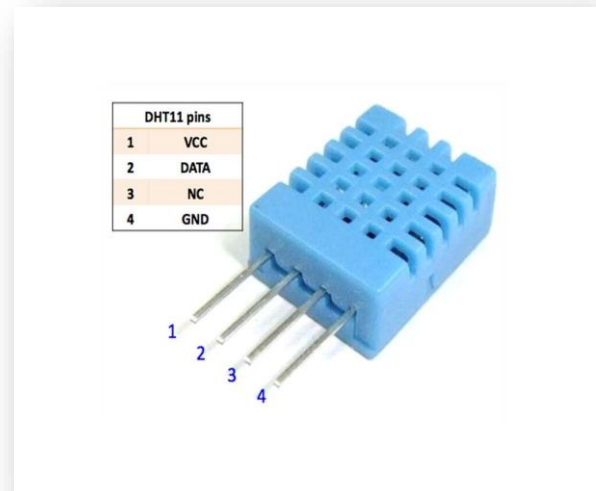
2.) Resistive touch sensor-

Resistive touch sensor measures touch through responding to the **pressure applied to their surface**. It consists of two conductive layers and a non-conductive separator. Unlike the capacitive touch sensors, it's not multi-touch compatible



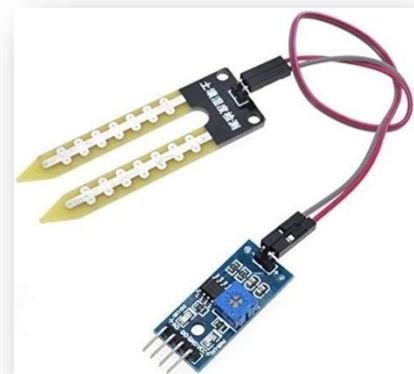
9. Humidity sensor (DHT11) :

Humidity is defined as the amount of water vapour in an atmosphere of air or other gases. These sensors usually follow the use of temperature sensors, as many manufacturing processes require perfect working conditions. Through measuring humidity, you can ensure that the whole process runs smoothly, and when there is any sudden change, action can be taken immediately, as sensors detect the change almost instantaneously. Their applications and use can be found in Industrial & residential domain for heating, ventilating, and air conditioning systems control



10. Soil Moisture sensor:

The Soil Moisture Sensor is used to measure the volumetric water content of soil. This makes it ideal for performing experiments in courses such as soil science, agricultural science, environmental science, horticulture, botany, and biology.



11. Temperature sensor (LM35) :

The **LM35** temperature sensor works by converting temperature into a proportional analog voltage output. This voltage is directly calibrated to the Celsius temperature scale, with a linear relationship of 10 mV per degree Celsius.

The formula used to convert the voltage into temperature is as given as - **Temp = ((5 × voltage × 100)/1024)**

